

1-(a) transposeMatrix

```
0. Add matrix
1. transposeMatrix
2. normalizeVector
3. calculateLength
4. scaleMatrix
5. multiplyTwoMatrices
6. addTwoMatrices
7. List all stored matrices
8.  $B = H^T A H$ ,  $C = H B H^T$ 
9. Exit
Choose: 1
A matrix name: A

A =
1.00 2.00
3.00 4.00

transpose =
1.00 3.00
2.00 4.00
```

1-(b) normalizeVector

```
0. Add matrix
1. transposeMatrix
2. normalizeVector
3. calculateLength
4. scaleMatrix
5. multiplyTwoMatrices
6. addTwoMatrices
7. List all stored matrices
8.  $B = H^T A H$ ,  $C = H B H^T$ 
9. Exit
Choose: 2
v matrix name: v

v =
3.00
2.00

normalize =
0.83
0.55
```

1-(c) calculateLength

```

0. Add matrix
1. transposeMatrix
2. normalizeVector
3. calculateLength
4. scaleMatrix
5. multiplyTwoMatrices
6. addTwoMatrices
7. List all stored matrices
8.  $B = H^T A H$ ,  $C = H B H^T$ 
9. Exit
Choose: 3
v matrix name: v
length(v) = 3.6056

```

1-(d) scaleMatrix

```

0. Add matrix
1. transposeMatrix
2. normalizeVector
3. calculateLength
4. scaleMatrix
5. multiplyTwoMatrices
6. addTwoMatrices
7. List all stored matrices
8.  $B = H^T A H$ ,  $C = H B H^T$ 
9. Exit
Choose: 4
A matrix name: A
scalar c: 3

A =
1.00 2.00
3.00 4.00

c*A =
3.00 6.00
9.00 12.00

```

1-(e) multiplyTwoMatrices

```

0. Add matrix
1. transposeMatrix
2. normalizeVector
3. calculateLength
4. scaleMatrix
5. multiplyTwoMatrices
6. addTwoMatrices
7. List all stored matrices
8.  $B = H^T A H$ ,  $C = H B H^T$ 
9. Exit
Choose: 5
A matrix name: A
B matrix name: H

A*B =
3.00 -1.00
7.00 -1.00

```

1-(f) addTwoMatrices

```
0. Add matrix
1. transposeMatrix
2. normalizeVector
3. calculateLength
4. scaleMatrix
5. multiplyTwoMatrices
6. addTwoMatrices
7. List all stored matrices
8.  $B = H^T A H$ ,  $C = H B H^T$ 
9. Exit
Choose: 6
A matrix name: A
B matrix name: H

A+B =
2.00 3.00
4.00 3.00
```

2-(b) $B = H^T A H$, $C = H B H^T$

```
0. Add matrix
1. transposeMatrix
2. normalizeVector
3. calculateLength
4. scaleMatrix
5. multiplyTwoMatrices
6. addTwoMatrices
7. List all stored matrices
8.  $B = H^T A H$ ,  $C = H B H^T$ 
9. Exit
Choose: 8
A matrix name: A
H_tilde matrix name: H

A =
1.00 2.00
3.00 4.00

H =
1.00 1.00
1.00 -1.00

H =
0.71 0.71
0.71 -0.71

B =  $H^T A H$  =
5.00 -1.00
-2.00 0.00

C =  $H B H^T$  =
1.00 2.00
3.00 4.00
```